Week 10 — Multiple regression I: one outcome, several causes

Business Forecasting

My code (.R) Notes

Back to Ecobici

Since W07, one regressor has forecast the daily trips:

$$\widehat{\text{trips}} = 16{,}893 + 723.6 \times \text{temp}, \qquad R^2 = 0.088$$

Temperature explains 8.8% of the day-to-day movement in trips. The fleet manager needs a better forecast.

What else moves the number of trips on a given day? Write down your guesses. For each one: can we measure it, and which way should it push trips?

From guesses to a regression

Take two guesses: days off and Fridays.

- The outcome y: trips that day.
- The regressors: temperature, a day off (0 or 1), a Friday (0 or 1).

The regression to write:

$$\text{trips}_i = \beta_0 + \beta_1 \, \text{temp}_i + \beta_2 \, \text{dayoff}_i + \beta_3 \, \text{friday}_i + u_i$$

Each β is the change in trips when only its own variable changes and the others are held fixed. OLS finds all four numbers at once.

Build the regression

Tick the regressors the class proposed. The regression to estimate appears first; then show the estimates.

What moves the number of Ecobici trips? (781 weekdays, 2017-2019)

REGRESSORS

- ☐ **`temp`** temperature, °C
- ☐ **`dayoff`** 1 on a holiday or the Christmas break
- ☐ **`monday`** 1 on Mondays
- ☐ **`friday`** 1 on Fridays
- ☐ **`pm25`** pollution, PM2.5 in µg/m^3

THE REGRESSION TO ESTIMATE

$\texttt{trips}_i = \beta_0 + u_i$

Show the estimates | Clear

Exercise 1 — A better forecast

`bikes` holds the 781 weekdays of W07–W09: `bikes$trips`, `bikes$temp`, `bikes$dayoff` (1 on a holiday or during the Christmas break) and `bikes$friday` (1 on Fridays).

How much better does the forecast get with days off and Fridays? How many trips should the fleet expect on a working Friday at 20 °C?

1. Regress `bikes$trips` on `bikes$temp`, `bikes$dayoff` and `bikes$friday`.
2. Compare R^2 and the residual standard error with temperature alone (0.088 and 5,302 trips).
3. Interpret the slope on `dayoff` and the slope on `friday`.
4. Forecast a working Friday at 20 °C.

Exercise | Start Over | Show Hint | Run Code

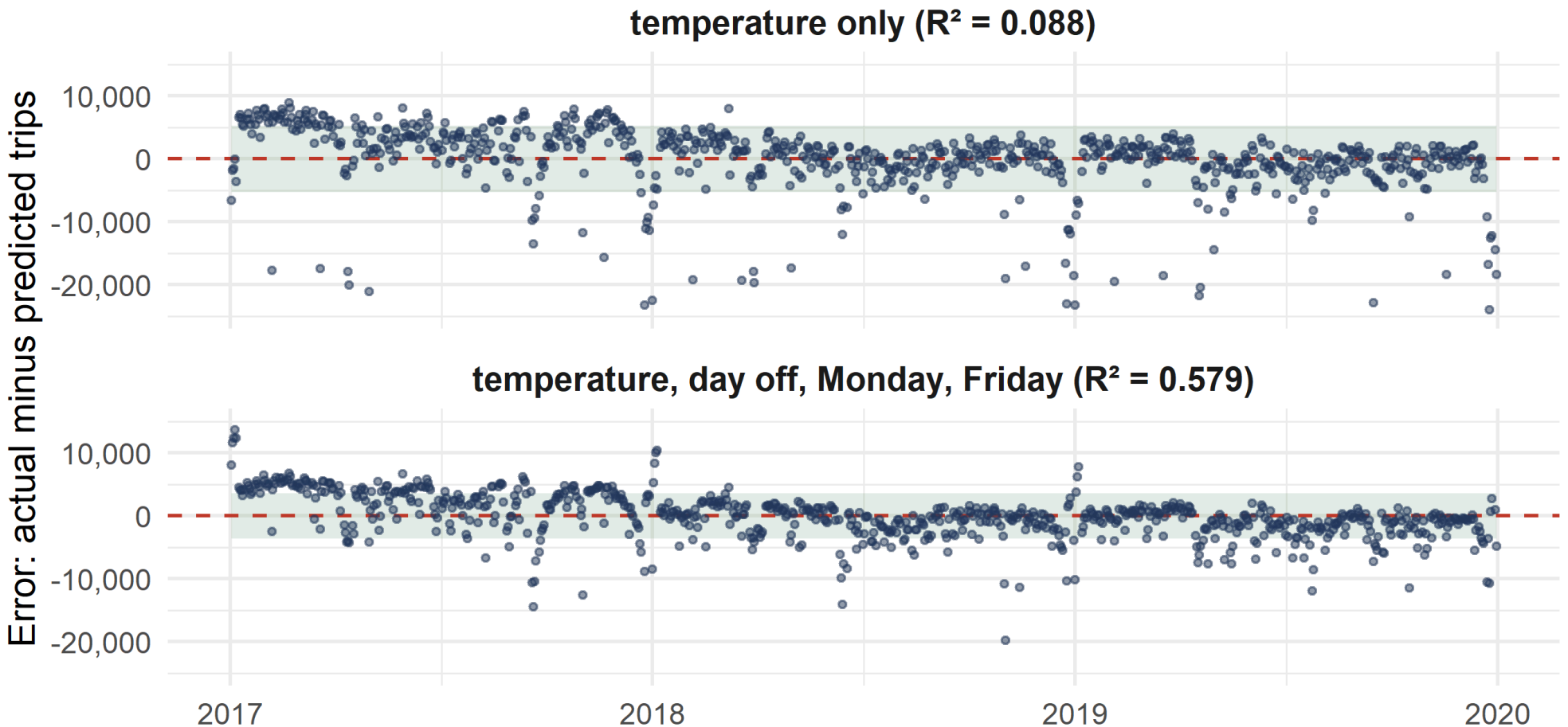
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How much better?

Regressors	R^2	adjusted R^2	Typical miss (trips)
temp	0.088	0.087	5,302
temp + dayoff	0.559	0.558	3,690
temp + dayoff + friday	0.569	0.567	3,651
temp + dayoff + monday + friday	0.579	0.577	3,609
temp + dayoff + monday + friday + pm25	0.579	0.577	3,611

- The days off do most of the work: R^2 jumps from 0.088 to 0.559.
- Mondays and Fridays add a little more. Pollution adds nothing: adjusted R^2 falls.
- A better regression is a better forecast: the typical miss shrinks from 5,302 to about 3,600 trips.

The errors, before and after



- Each dot is one day's error. The green band is plus or minus one typical miss.
- With temperature only, the holidays and the Christmas breaks sit far below zero. Once the days off are in the model, those big misses are gone and the band is narrower.

Regression at work

The same steps work for any business question: name the outcome, list what moves it, write the regression, estimate it, read each slope with the others held fixed.

Business question	Data
What is an extra bedroom worth?	88 house sales
How much is a year of education worth? Is there a gender gap?	526 workers
Do fast-food chains charge more for a soda in richer neighbourhoods?	401 restaurants
What does a home run earn a baseball player?	353 players
Does a CEO's pay follow the firm's results?	209 CEOs

All five come from the `wooldridge` package: `hprice1`, `wage1`, `discrim`, `mlb1`, `ceosal1`.

What is an extra bedroom worth?

- The outcome: the sale price of a house, in thousands of dollars.
- What moves it? Size, bedrooms, the lot, the style.

$$\text{price}_i = \beta_0 + \beta_1\,\text{bdrms}_i + \beta_2\,\text{sqrft}_i + \beta_3\,\text{lotsize}_i + \beta_4\,\text{colonial}_i + u_i$$

β_1 is what a bedroom adds for a house of the same size, lot and style. Is that different from comparing houses with more and fewer bedrooms?

How much is a year of education worth?

- The outcome: hourly wage, in dollars.
- What moves it? Education, experience, years with the employer, sex, marriage.

$$\text{wage}_i = \beta_0 + \beta_1\,\text{educ}_i + \beta_2\,\text{exper}_i + \beta_3\,\text{tenure}_i + \beta_4\,\text{female}_i + u_i$$

β_4 is the gap between a woman and a man with the same education, experience and tenure: the part of the pay gap these three cannot explain.

Do richer neighbourhoods pay more for a soda?

- The outcome: the price of a small soda at a fast-food restaurant, in dollars.
- What moves it? Local income, poverty, competing stores, the chain.

$$\text{psoda}_i = \beta_0 + \beta_1 \,\text{income}_i + \beta_2 \,\text{prppov}_i + \beta_3 \,\text{nstores}_i + \beta_4 \,\text{BK}_i + \beta_5 \,\text{KFC}_i + u_i$$

β_1 asks whether a restaurant charges more where incomes are higher, comparing restaurants of the same chain facing the same competition.

What does a home run earn?

- The outcome: a player's salary, in thousands of dollars a year.
- What moves it? Years in the league, games played, batting average, home runs, runs batted in.

$$\text{salary}_i = \beta_0 + \beta_1 \, \text{years}_i + \beta_2 \, \text{gamesyr}_i + \beta_3 \, \text{bavg}_i + \beta_4 \, \text{hrunsyr}_i + \beta_5 \, \text{rbisyr}_i + u_i$$

β_4 is what one more home run a year is worth for a player with the same experience and the same other statistics.

Does a CEO's pay follow the firm's results?

- The outcome: the CEO's salary, in thousands of dollars.
- What moves it? The firm's return on equity, its sales, its stock return.

$$\text{salary}_i = \beta_0 + \beta_1 \, \text{roe}_i + \beta_2 \, \text{sales}_i + \beta_3 \, \text{ros}_i + u_i$$

β_1 asks whether a better-run firm pays its CEO more, comparing firms of the same size.

Estimate them

Pick a question, tick the regressors, and compare the slopes and R^2 across models.

Houses: sale price

What is an extra bedroom worth? (88 house sales, Boston area, 1990 (wooldridge: hprice1))

REGRESSORS

- ☐ **bdrms** bedrooms
- ☐ **sqrft** size, square feet
- ☐ **lotsize** lot, square feet
- ☐ **colonial** 1 if colonial style

THE REGRESSION TO ESTIMATE

$\text{price}_i = \beta_0 + u_i$

Show the estimates | Clear

Exercise 2 — What is an extra bedroom worth?

`houses` holds 88 house sales from the `wooldridge` package (`hprice1`):

- `houses$price` — sale price, thousands of dollars
- `houses$bdrms` — number of bedrooms
- `houses$sqrft` — size of the house, square feet

1. Regress `houses$price` on `houses$bdrms`. How much is one more bedroom worth?
2. Regress `houses$price` on `houses$bdrms` and `houses$sqrft`. How much is one more bedroom worth now, for a house of the same size?
3. Compute the correlation between `houses$bdrms` and `houses$sqrft`. Why did the bedroom slope fall?

Exercise | Start Over | Show Hint | Run Code

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1
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Holding size fixed

Model	Bedroom slope	Size slope	R^2
price on bedrooms	62.0		0.258
price on bedrooms and size	15.2	0.128	0.632

- An appraiser who prices bedrooms alone pays for size twice.
- With size held fixed, an extra bedroom is a smaller house cut into more rooms.

The rest of this week asks the same question of Mexican data from October 2025: how late will a flight land, once the other causes are held fixed?

How late will this flight land?

You plan the day for an airline at Mexico City airport (AICM). A flight from Monterrey left 40 minutes late. The gate, the ground crew and the passengers with a connection all need one number: how many minutes late will it land?

What moves the arrival delay? Write down your guesses. For each one: can we measure it, and which way should it push the delay?

Each cause needs its own number, measured with the others held fixed.

The data: arrivals at AICM

`flights` holds 11,887 flights that landed at Mexico City airport (AICM) in October 2025, from 111 airports. One row is one flight.

Column	What it is
`arr_delay`	minutes late at arrival (negative = early)
`dep_delay`	minutes late at departure (negative = early)
`sched_time`	scheduled flight time, minutes
`origin`, `code`, `country`, `intl`	where the flight comes from; `intl` is 1 for a flight from outside Mexico
`airline`	the airline
`date`, `weekday`, `hour`	day and hour of the scheduled arrival
`rain`, `wind`, `temp`	weather at AICM in that hour: rain (mm), wind (km/h), temperature (°C)

Schedules include slack, so most flights land before the scheduled time: only 14% land late.

Sources: arrival records of a public flight tracker; weather from the Open-Meteo historical archive. 359 records with extreme or impossible times are left out.

1 · Departure and arrival

One regressor first: how much of a late departure is still there at landing?

Exercise 3 — Late out, late in

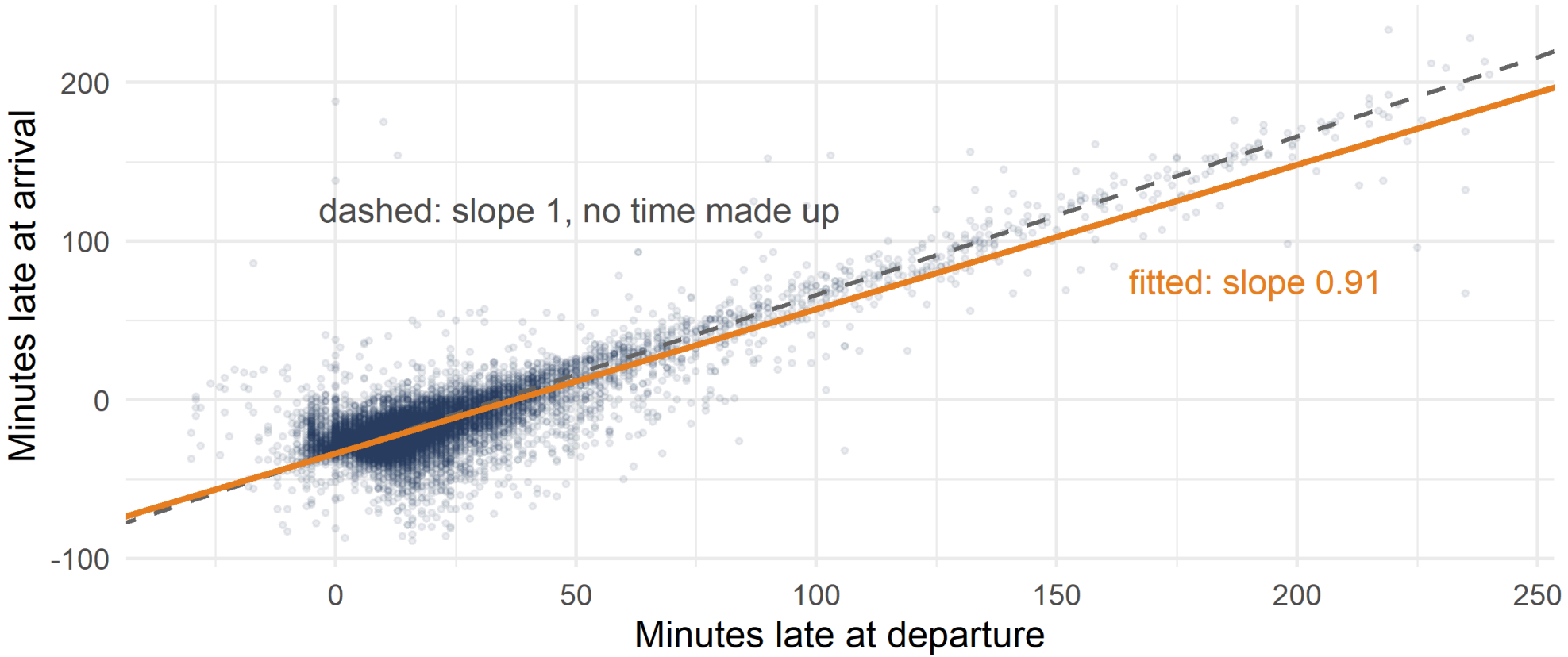
How many minutes late does a flight land for each minute it leaves late? How early does a flight land if it leaves on time?

1. Regress `flights$arr_delay` on `flights$dep_delay`.
2. Interpret the slope and the intercept, in minutes.
3. How much of the variation in arrival delays does the departure delay explain?

Exercise | Start Over | Show Solution | Run Code

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1
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Most of a late departure stays



- The dashed line is a flight that makes up no time. The orange line, the fit, sits a little below it: about 5 seconds made up per minute lost.
- $R^2 = 0.755$. The rest is the route, the airline, the weather and luck.

2 · A second regressor

Two causes in one model, and what “held fixed” means.

Do long flights make up more time?

A long flight has more time in the air, and more slack in its schedule, to make up a late departure.

$$\text{arr_delay}_i = \beta_0 + \beta_1 \, \text{dep_delay}_i + \beta_2 \, \text{sched_time}_i + u_i$$

- `sched_time` runs from 40 minutes to almost 16 hours. From Monterrey it is about 100 minutes; from Madrid about 720.
- β_2 compares two flights that left equally late, one with a longer schedule.

Exercise 4 — Does a long flight make up time?

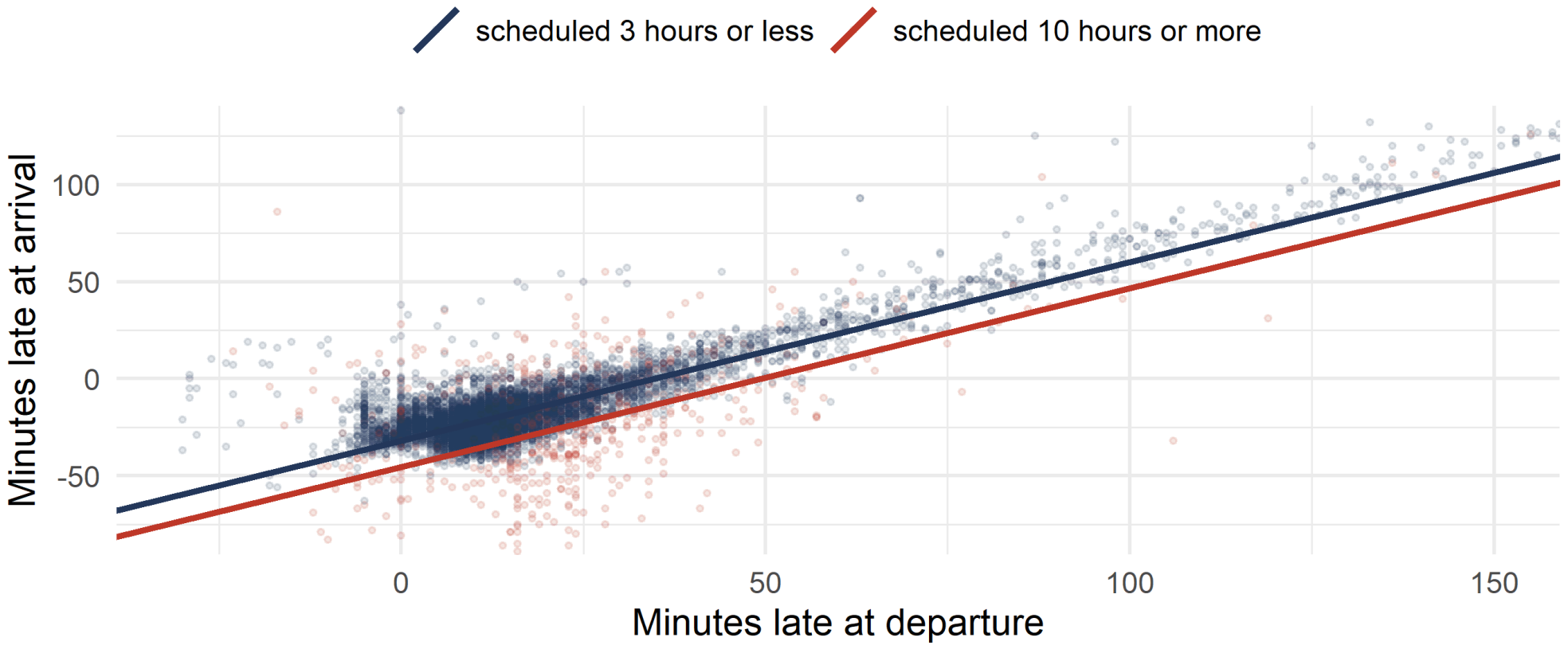
For two flights that left equally late, does the one with the longer schedule land earlier? By how many minutes per hour of schedule?

1. Regress `flights$arr_delay` on `flights$dep_delay` and `flights$sched_time`.
2. Interpret both slopes, in minutes. Convert the `sched_time` slope to one extra hour of scheduled flight.
3. Did the `dep_delay` slope move from Exercise 3? How much did R^2 rise?

Exercise | Start Over | Show Hint | Run Code

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1
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Two flights, two lines



- Two lines with the same slope, 0.92, drawn at 100 scheduled minutes (Monterrey) and 720 (Madrid). They are 13.5 minutes apart: 620 minutes $\times$ 0.0217.
- Holding the scheduled time fixed means moving along one line.
- A 0/1 regressor, like `dayoff` for Ecobici, works the same way with only two values: it moves the whole line up or down.

Exercise 5 — Two forecasts

Two flights leave 40 minutes late: one from Monterrey, scheduled for 100 minutes, and one from Madrid, scheduled for 720 minutes. The gate team needs the expected arrival delay of each.

1. Compute both forecasts by hand from the coefficients of the regression of `flights$arr_delay` on `flights$dep_delay` and `flights$sched_time`.
2. Compute them again with `predict()`.

Exercise | Start Over | Show Hint | Run Code

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1
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3 · When a new regressor moves a slope

Adding the scheduled time barely moved the departure slope. When does a new regressor move one?

What the schedule planner sees

A schedule planner looks at the scheduled time only: do longer flights land earlier?

Predict before you fit: what happens to the `sched_time` slope once the departure delay is held fixed?

Exercise 6 — Is the slack hidden?

1. Regress `flights$arr_delay` on `flights$sched_time` alone.
2. Compare its slope with the `sched_time` slope of Exercise 4, where `flights$dep_delay` is held fixed.
3. Compute the correlation between `flights$sched_time` and `flights$dep_delay`.
4. Why did the `sched_time` slope move? What does it mean for the planner?

Exercise | Start Over | Show Solution | Run Code

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The rule

New regressor	Correlation with sched_time	Moves the arrival delay?	sched_time slope
rain at AICM	0.01	no: -0.25 minutes per mm, not significant	-0.0069 → -0.0069
departure delay	0.09	yes: 0.92 minutes per minute	-0.0069 → -0.0217

A slope moves only when the new regressor does both: it moves the outcome, and it moves with the old regressor. Leaving such a variable out is omitted-variable bias.

A small correlation is enough when the left-out cause is strong: the departure delay moves the arrival almost minute for minute.

The algebra behind it

4 · How OLS finds the coefficients

The same rule as W07, the smallest sum of squared residuals, written for many regressors.

Still least squares

$$SSE(b_0, b_1, b_2) = \sum_{i=1}^{n} (y_i - b_0 - b_1 x_{i1} - b_2 x_{i2})^2$$

- OLS picks the three numbers that make SSE as small as possible.
- With one regressor this gave the W07 formulas for $\hat{\beta}_0$ and $\hat{\beta}_1$.
- With several, the answer is easiest to write with matrices.

The regression in matrix form

$$\underbrace{\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}}_{\mathbf{y}:\, n\times 1} = \underbrace{\begin{bmatrix} 1 & x_{11} & \cdots & x_{1k} \\ 1 & x_{21} & \cdots & x_{2k} \\ \vdots & \vdots & & \vdots \\ 1 & x_{n1} & \cdots & x_{nk} \end{bmatrix}}_{\mathbf{X}:\, n\times(k+1)} \underbrace{\begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_k \end{bmatrix}}_{\boldsymbol{\beta}:\, (k+1)\times 1} + \underbrace{\begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix}}_{\mathbf{u}:\, n\times 1}$$

The OLS estimator, on your formula sheet:

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$$

`lm()` does this for you. A version with four flights, by hand, is in the appendix.

Four flights, by hand | See the matrices move

Full rank: no column can be a copy

- $\mathbf{X}$ must have full rank: no column can be built exactly from the others.
- If one can, $(\mathbf{X}^\top \mathbf{X})^{-1}$ does not exist, and OLS has no single answer.

A colleague adds the scheduled time in hours, next to the scheduled time in minutes. What happens if both go into the same regression?

Exercise 7 — Minutes and hours

1. Add `flights$sched_hours`, the scheduled time in hours: `flights$sched_time / 60`.
2. Regress `flights$arr_delay` on `flights$dep_delay`, `flights$sched_time` and `flights$sched_hours`. What does `summary()` print for `sched_hours`? Why?

Exercise | Start Over | Show Solution | Run Code

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1
```

5 · Collinearity

Near-copies are allowed, but they make each slope hard to pin down.

Today or yesterday?

Back to Ecobici. `bikes$temp_yday` is yesterday's temperature. Riders may decide on today's weather, or on what they remember from yesterday.

Today's and yesterday's temperature move together. What happens to the temperature slope, and to its standard error, when both go into the regression?

Exercise 8 — Today or yesterday?

1. Regress `bikes$trips` on `bikes$temp` and `bikes$dayoff`.
2. Add `bikes$temp_yday`.
3. Compare the slope on `temp` and its standard error in the two regressions. What happened to R^2?
4. Compute the correlation between `bikes$temp` and `bikes$temp_yday`. The first day has no yesterday, so use `use = "complete.obs"`.

Exercise | Start Over | Show Solution | Run Code

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1
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Collinearity costs precision

Model	temp slope	SE of temp slope	R^2
trips on temp and dayoff	395	59	0.559
trips on temp, temp_yday and dayoff	525	90	0.562

- Today's and yesterday's temperature move together (correlation 0.76). The data can hardly tell which one riders respond to: the standard error of the temperature slope grows by half, and yesterday's slope even takes the wrong sign.
- The fit barely changes: R^2 goes from 0.559 to 0.562.
- The hours column of Exercise 7 is the extreme: a perfect copy, and R drops it.

For a forecast, collinearity costs little. For saying which cause matters, it costs a lot.

6 · Fit with several regressors

More regressors always fit the sample at least as well. Is the gain worth it?

Does the weather matter?

One route only: the 682 flights from Guadalajara, all scheduled for about 90 minutes.

Once the departure delay is known, does the weather at AICM help forecast the arrival delay?

Exercise 9 — The weather and adjusted R^2

1. Keep the Guadalajara flights: `gdl <- flights[flights$code == "GDL", ]`. Regress `gdl$arr_delay` on `gdl$dep_delay`. Note the R^2 and the adjusted R^2.
2. Add `gdl$rain`. Note both again. Is `rain` significant at 5%?
3. Replace `rain` with `gdl$wind`. Note both again.
4. Which weather variable raised the adjusted R^2?

Exercise | Start Over | Show Solution | Run Code

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1
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R^2 against adjusted R^2

$$R^2 = 1 - \frac{SSE}{SST} \qquad \bar{R}^2 = 1 - \frac{SSE/(n-k-1)}{SST/(n-1)}$$

- A new regressor can only lower SSE, so R^2 can only rise, even for a useless variable.
- Adjusted R^2 charges one degree of freedom per regressor. It rises only when the new regressor's $|T|$ is above 1.
- To compare models with different numbers of regressors, use adjusted R^2.

For the operations team: the departure delay carries the forecast, and a long schedule absorbs part of it. The weather at the airport adds almost nothing once the departure delay is known.

Do some airlines make up more time than others? That needs a regressor with several categories: week 11.

Exercise 10 — Brief the operations manager

Give the manager the expected arrival delay of flights that leave 0, 30 and 60 minutes late, from Monterrey (100 scheduled minutes) and from Madrid (720), with one picture and one sentence.

1. Use the regression of `flights$arr_delay` on `flights$dep_delay` and `flights$sched_time`.
2. Forecast the six delays with `predict()`.
3. Plot `flights$arr_delay` against `flights$dep_delay`.
4. Write one sentence for the manager.

Exercise | Start Over | Show Hint | Run Code

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1
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What you should be able to do

- Explain why one cause at a time misleads, and what “held fixed” means.
- Fit a regression with several regressors and read each slope with the others held fixed.
- Read the coefficient of a 0/1 regressor as the gap between two parallel lines.
- Forecast with several regressors, by hand and with `predict()`.
- Say when adding a regressor moves another slope, and why.
- Explain full rank, and what R does with an exact copy of a regressor.
- Recognise collinearity in the standard errors.
- Compare models with adjusted R^2.

Next week: regressors with several categories, such as the airline, and interactions, with PROFECO shelf prices: whether small local stores price to their neighbourhood, and what the January 2026 tax did to drink prices.

Go to the exam practice →

Practice for this week

Weekly exercise set: `W10_exercises.pdf`, with solutions in `W10_solutions.pdf`.

You can already solve exercises 2, 3 and 25, and most of 4, 5 and 6.

Not in these slides: 1 and 7(c) use regressors with several categories (week 11); 4(f), 5(i) and 6(b) use intervals for a coefficient or a forecast (week 12); 7(a)–(b) use a squared regressor, which this course does not cover.

Suggested before the next class, closest to the exam:

Exercise	Topic	How
2	Grades on two GRE scores and a 0/1 regressor (`metricsgrades`)	R with the data
3	Cigarette tax and price move together (`cigdata`)	R with the data
25	Food spending on income and household size (ENIGH)	R with the data
5 (a)–(h)	Vote shares on four district traits (`congress`)	R with the data

Interactive versions: exam practice

Exam practice

Problems from this week's set. ← Back to the summary · ← Back to Practice for this week

Exam practice 1 — Grades and the GRE

[P3 #2]

`metricsgrades` holds 68 students of a graduate econometrics course, from the `probstats4econ` package.

- `metricsgrades$total` — final course grade, out of 100
- `metricsgrades$gre_quant`, `metricsgrades$gre_verbal` — GRE scores, out of 170
- `metricsgrades$domestic` — 1 if a US student, 0 otherwise

a. Compute the correlation matrix of the four variables. Which one has the largest correlation, in magnitude, with `total`?

b. Regress `total` on the other three variables.

c. Interpret the slope on `gre_quant`.

d. Predict the grade of a non-domestic student with 160 in quant and 150 in verbal.

e. What is the estimated standard deviation of the errors?

f. Test $H_0 : \beta_{\text{domestic}} = 0$ against $H_A : \beta_{\text{domestic}} \neq 0$ at the 10% level, using the output of `summary()`.

g. Drop `domestic` and fit again. What changes?

h. Put `domestic` back and drop `gre_verbal`. What happens to the significance of `domestic`, and why?

Exercise | Start Over | Show Solution | Run Code

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Exam practice 2 — Cigarette taxes and prices

[P3 #3]

`cigdata` holds the 50 US states and Washington D.C. in 2019, from the `probstats4econ` package.

- `cigdata$cigsales` — packs sold per person per year
- `cigdata$cigtax` — state tax per pack, dollars
- `cigdata$cigprice` — total price per pack, dollars
- `cigdata$producer` — 1 if the state grows tobacco

a. Regress `cigsales` on `cigtax` and `producer`. Interpret the slopes and R^2.

b. Add `cigprice`. How does R^2 change?

c. What happens to the significance of the tax slope?

d. Use the correlation between `cigtax` and `cigprice` to explain (c).

e. Drop `cigtax` and keep `cigprice` and `producer`. Compare with (a) and (b).

f. Which model do you prefer, (`cigtax`, `producer`) or (`cigprice`, `producer`)? Why?

Exercise | Start Over | Show Solution | Run Code

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1
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Exam practice 3 — Mutual fund returns

[P3 #4]

`mutualfunds` holds 208 large-blend equity funds, from the `probstats4econ` package.

- `mutualfunds$return_10yr` — ten-year annualised return (0.08 = 8%)
- `mutualfunds$expense_ratio` — annual fee (0.005 = 0.5%)
- `mutualfunds$manager_tenure`, `mutualfunds$fund_age` — years
- `mutualfunds$load` — "Y" if the fund charges a sales fee, "N" otherwise

a. Add a column `hasload`, 1 if `load` is "Y" and 0 otherwise.

b. Regress `return_10yr` on `expense_ratio`, `manager_tenure`, `fund_age` and `hasload`.

c. Interpret R^2.

d. What does a 0.001 rise in `expense_ratio` do to the ten-year return?

e. Which slopes are significant at 5%?

Exercise | Start Over | Show Solution | Run Code

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Exam practice 4 — District elections

[P3 #5]

`congress` holds the 476 US House district elections of 1990, from the `probstats4econ` package.

- `congress$demvoteshare` — share of the vote for the Democrat (0 to 1)
- `congress$democrat` — 1 if the Democrat won; `congress$lagdemocrat` — 1 if they won the election before
- `congress$medianincome` — median district income, dollars
- `congress$pcturban`, `congress$pctblack`, `congress$pcthighschl` — shares urban, Black, with a high-school degree

a. Draw a histogram of `demvoteshare`. One peak or two?

b. What fraction of the elections did Democrats win?

c. Compute the mean of each of the four district traits where Democrats won and where Republicans won.

d. For each trait, regress it on `democrat` and say, from the p-value, whether its mean differs between the two groups at the 5% level.

e. Regress `demvoteshare` on the four traits.

f. Interpret the income slope for a 1,000-dollar rise.

g. Interpret the urban slope for a rise of 10 percentage points.

h. Add `lagdemocrat`. What happens to R^2 and to the other slopes?

Exercise | Start Over | Show Solution | Run Code

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1
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Exam practice 5 — Out-of-pocket medical costs

[P3 #6]

`hrs` holds 6,052 unmarried people aged 50 and over, from the Health and Retirement Study (`probstats4econ` package). Use only the 3,983 with positive out-of-pocket costs.

- `hrs$outofpocket_costs` — medical costs paid out of pocket in 2000, dollars
- `hrs$age` — years
- `hrs$ins_none` — 1 if no health insurance; `hrs$ins_medicare` — 1 if Medicare
- `hrs$male` — 1 if male

a. Regress `outofpocket_costs` on `age`, `ins_none`, `ins_medicare` and `male`. Interpret the slopes on `age` and `ins_none`.

Exercise | Start Over | Show Solution | Run Code

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1
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📝 Exam practice 6 — A second regressor for food spending

[25]

`Hp` holds the 74,639 ENIGH households with positive income (Canvas, 02 Data, `Dataset_spending_income.Rda`).

- `Hp$food_spending` — monthly food spending, pesos
- `Hp$monthly_income` — monthly income, pesos
- `Hp$HHmembers` — people in the household

a. Regress `food_spending` on `monthly_income`. Report the slope and R^2.

b. Add `HHmembers`. Report the three coefficients, the two standard errors and R^2.

c. Interpret the slope on `HHmembers`.

d. How does the meaning of the income slope differ between (a) and (b)?

e. The income slope barely moved. Use the correlation between income and household size to explain why.

f. A supermarket chain sizes its stores from these two variables. Which matters more for food spending, and what is the evidence?

Exercise | Start Over | Show Solution | Run Code

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1
```

Appendix — optional

The algebra of omitted-variable bias, and the matrix formula by hand.

Appendix — the algebra of omitted-variable bias

True model: $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + u$. We fit y on x_1 alone.

$$x_2 = \delta_0 + \delta_1 x_1 + v$$

$$y = \beta_0 + \beta_1 x_1 + \beta_2 (\delta_0 + \delta_1 x_1 + v) + u$$

$$y = (\beta_0 + \beta_2 \delta_0) + (\beta_1 + \beta_2 \delta_1)\, x_1 + (\beta_2 v + u)$$

$$\text{slope alone} = \boxed{\beta_1 + \beta_2\, \delta_1}$$

Start
Write the left-out regressor as a line in x_1. δ_1 says how they move together.

Plug it in
Replace x_2 in the true model.

Collect the terms in x_1
The slope on x_1 alone is the red part.

Both conditions
The bias $\beta_2 \delta_1$ is zero if x_2 does not move y ($\beta_2 = 0$) or does not move with x_1 ($\delta_1 = 0$).

In the flights data, with x_1 = `sched_time` and x_2 = `dep_delay`, the identity holds exactly for the OLS estimates: $-0.0217 + 0.9215 \times 0.0161 = -0.0069$.

← Back to the rule

Appendix — four flights, by hand

Four flights: minutes late at departure, scheduled time in hours, and minutes late at arrival.

Flight	Departure delay x_1 (minutes)	Scheduled time x_2 (hours)	Arrival delay y (minutes)
1	0	1	−22
2	10	1	−16
3	10	2	−20
4	20	2	−6

$$\mathbf{X} = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 10 & 1 \\ 1 & 10 & 2 \\ 1 & 20 & 2 \end{bmatrix} \qquad \mathbf{y} = \begin{bmatrix} -22 \\ -16 \\ -20 \\ -6 \end{bmatrix}$$

The blue column of 1s carries the intercept.

← Back to the matrix form

Appendix — four flights: the steps

$$\mathbf{X}^\top\mathbf{X} = \begin{bmatrix} 4 & 40 & 6 \\ 40 & 600 & 70 \\ 6 & 70 & 10 \end{bmatrix}$$

$$(\mathbf{X}^\top\mathbf{X})^{-1} = \begin{bmatrix} 2.75 & 0.05 & -2 \\ 0.05 & 0.01 & -0.1 \\ -2 & -0.1 & 2 \end{bmatrix}$$

$$\mathbf{X}^\top\mathbf{y} = \begin{bmatrix} -64 \\ -480 \\ -90 \end{bmatrix}$$

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{y} = \boxed{\begin{bmatrix} -20 \\ 1 \\ -4 \end{bmatrix}}$$

How the regressors move together
Row by column: $n = 4$, $\sum x_1 = 40$, $\sum x_1^2 = 600$, $\sum x_2 = 6$, $\sum x_1 x_2 = 70$, $\sum x_2^2 = 10$.

Untangle them
The inverse: multiplied by $\mathbf{X}^\top\mathbf{X}$ it gives the identity matrix.

How each regressor moves with y
$\sum y = -64$, $\sum x_1 y = -480$, $\sum x_2 y = -90$.

Multiply
One minute late at arrival per minute late at departure, for the same scheduled time; 4 minutes earlier per extra scheduled hour, for the same departure delay. The residuals are ± 2.

← Back to the matrix form